

Extra Notes on Direct Limits

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February 24, 2026

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Here, we'll fix I as a directed set (a preorder + every two elements has an upper bound).

1 Abelian Category Structure of $[I, \mathbf{Ab}]$

Recall that for a directed system in $\mathbf{Ab}$, one can take its direct limit. However, if fixing a direct system I , and consider the functor category $[I, \mathbf{Ab}]$ (sometimes even stronger, $[I, \mathbf{CRing}]$), it has a natural abelian category structure:

Proposition

$[I, \mathbf{Ab}]$ has a natural Abelian Category structure.

Proof:

1. Zero Object:

If taking the „zero system“ $z : I \rightarrow \mathbf{Ab}$ by $z(i) = 0$, and $zf_{ij} : z(i) \rightarrow z(j)$ to be the zero morphism, then for any other direct system $F : I \rightarrow \mathbf{Ab}$, the only possible natural transformation from z , say $\mu : z \rightarrow F$, is $\mu(i) = 0 : z(i) = 0 \rightarrow F(i)$; and the only possible natural transformation to z , say $\rho : F \rightarrow z$, is $\rho(i) = 0 : F(i) \rightarrow z(i) = 0$ (since 0 the trivial group is both initial and final in $\mathbf{Ab}$).

Notice that this automatically makes μ, ρ as natural transformations, as these two diagrams commute:

$$\begin{array}{ccc} z(i) & \xrightarrow{0} & z(j) \\ \mu_i=0 \downarrow & & \downarrow \mu_j=0 \\ F(i) & \xrightarrow{f_{ij}} & F(j) \\ & f_{ij} & \\ F(i) & \xrightarrow{f_{ij}} & F(j) \\ \rho_i=0 \downarrow & & \downarrow \rho_j=0 \\ z(i) & \xrightarrow{0} & z(j) \\ & 0 & \end{array}$$

So, we indeed have a zero object in this category.

2. Kernels and Cokernels:

If given a morphism $\varphi : F \rightarrow G$ between two directed systems, for each $i \in I$, there exists a group homomorphism $\varphi_i : F(i) \rightarrow G(i)$, and hence one can define $K(i)$ as the set-theoretic kernel of $F(i)$, and the morphism $\ker(\varphi)_i : K(i) \hookrightarrow F(i)$ as the inclusion.

for the morphism $f_{ij} : F(i) \rightarrow F(j)$ and $g_{ij} : G(i) \rightarrow G(j)$, notice we have the following naturality square, regarding the kernels:

$$\begin{array}{ccccc} K(i) & \xrightarrow{\ker(\varphi)_i} & F(i) & \xrightarrow{\varphi_i} & G(i) \\ & & \downarrow f_{ij} & & \downarrow g_{ij} \\ K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j) \end{array}$$

Which, notice that one has $0 = g_{ij} \circ (\varphi_i \circ \ker(\varphi)_i) = \varphi_j \circ (f_{ij} \circ \ker(\varphi)_i)$, so the morphism $f_{ij} \circ \ker(\varphi)_i$ uniquely factors through $\ker(\varphi)_j : K(j) \hookrightarrow F(j)$, say some $k_{ij} : K(i) \rightarrow K(j)$:

$$\begin{array}{ccccc} K(i) & \xrightarrow{\ker(\varphi)_i} & F(i) & \xrightarrow{\varphi_i} & G(i) \\ \downarrow k_{ij} & & \downarrow f_{ij} & & \downarrow g_{ij} \\ K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j) \end{array}$$

Note the universality of kernels make $K : I \rightarrow F$ by $i \mapsto K(i)$ and $K(i \rightarrow j) = k_{ij}$ as a functor (or a directed system over I).

Now, we claim that K together with the morphism $\ker(\varphi) : K \rightarrow F$ is indeed a kernel of $\varphi : F \rightarrow G$: Suppose some other morphism $\psi : L \rightarrow F$ satisfies $\varphi \circ \psi = 0$, then each index $i \in I$ satisfies $\varphi_i \circ \psi_i : L(i) \rightarrow F(i) \rightarrow G(i)$ being 0. So, the map uniquely factors through $\ker(\varphi)_i : K(i) \rightarrow F(i)$, say as $\bar{\psi}_i : L(i) \rightarrow K(i)$, as follow:

$$\begin{array}{ccccccc} & & K(i) & \xrightarrow{\ker(\varphi)_i} & F(i) & \xrightarrow{\varphi_i} & G(i) \\ & & \searrow k_{ij} & & \searrow f_{ij} & & \searrow g_{ij} \\ \exists! \bar{\psi}_i \uparrow & & & & & & \\ & & K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j) \\ & \nearrow \psi_i & & \nearrow \ker(\varphi)_j & & & \\ L(i) & \xrightarrow{\psi_i} & K(j) & \xrightarrow{\ker(\varphi)_j} & F(j) & \xrightarrow{\varphi_j} & G(j) \\ & \searrow l_{ij} & & \searrow \psi_j & & & \\ & & L(j) & & & & \end{array}$$

Which, the square between $L(i), L(j), K(i), K(j)$ is a naturality square, because of the following diagram chase:

$$\begin{aligned} \ker(\varphi)_j \circ (k_{ij} \circ \bar{\psi}_i) &= f_{ij} \circ (\ker(\varphi)_i \circ \bar{\psi}_i) = f_{ij} \circ \psi_i \\ &= \psi_j \circ l_{ij} = \ker(\varphi)_j \circ (\bar{\psi}_j \circ l_{ij}) \end{aligned} \tag{1.1}$$

Finally, using the fact that $\ker(\varphi)_j$ is a kernel in $\mathbf{Ab}$ (hence a monomorphism), this concludes that $k_{ij} \circ \bar{\psi}_i = \bar{\psi}_j \circ l_{ij}$, hence it's a naturality square. With i, j being arbitrary, one concludes that $\bar{\psi} : L \rightarrow K$ is a natural transformation (with each component being $\bar{\psi}_i$).

The uniqueness of $\bar{\psi}$ (as a morphism in $[I, \mathbf{Ab}]$) follows the universality of each $\ker(\varphi)_i$ as a kernel in $\mathbf{Ab}$, proving that $\ker(\varphi) : K \rightarrow F$ is indeed a kernel in $[I, \mathbf{Ab}]$.

Note: For the cokernel case, just put the arrows on the other side, the proof works similarly.

3. Equivalence of Monomorphism and Component-Wise Monomorphism:

(**Note:** The dual notion is epimorphism and component-wise epimorphism, where the proof follows similarly.)

It's clear that if a morphism $\varphi : F \rightarrow G$ is component-wise monomorphic, it must be a monomorphism in $[I, \mathbf{Ab}]$ (just because the left cancellative property applies to each component will prove the equality globally).

Now, suppose $\varphi : F \rightarrow G$ is a monomorphism in $[I, \mathbf{Ab}]$ (i.e. given any two morphism $\rho, \mu : L \rightarrow F$, one has $\varphi \circ \rho = \varphi \circ \mu$ implies $\rho = \mu$), then we claim the kernel of each component is 0 (which is equivalent to being monomorphic in $\mathbf{Ab}$). Note that part 2 above helped us realize the kernel of the morphism as component-wise kernel. So, if consider the zero morphism $0 : K \rightarrow F$ and the kernel morphism $\ker(\varphi) : K \rightarrow F$, notice that $\varphi \circ 0 = \varphi \circ \ker(\varphi)$, which the left cancellative property implies $0 = \ker(\varphi)$, or each component has $0 = \ker(\varphi)_i : K(i) \rightarrow F(i)$, proving each $\varphi_i : F(i) \rightarrow G(i)$ is a monomorphism.

4. Equivalence of Monomorphism and Kernel:

(**Note:** The dual notion is epimorphism and cokernel, the proof also follows similarly.)

Since kernel of a morphism is a component-wise kernel (cf. part 2), and kernel is the same as monomorphism in $\mathbf{Ab}$, then part 3 guarantees the kernel morphism in $[I, \mathbf{Ab}]$ to be a monomorphism (as it's component-wise monomorphic).

Conversely, given a monomorphism $\varphi : F \hookrightarrow G$, running through similar proof of part 2 guarantees it has a cokernel $\text{coker}(\varphi) : G \rightarrow C$ (where it's taken as cokernel component-wise). Then, since for every component $\varphi_i : F(i) \rightarrow G(i)$ is a kernel of $\text{coker}(\varphi)_i : G(i) \rightarrow C(i)$ (in $\mathbf{Ab}$, a monomorphism is a kernel of its own cokernel), hence globally $\varphi : F \hookrightarrow G$ is a kernel of $\text{coker}(\varphi) : G \rightarrow C$, proving that φ is a kernel in $[I, \mathbf{Ab}]$.

5. Existence of Product, Coproduct, and Finite Biproduct:

Given two direct systems $F, G \in [I, \mathbf{Ab}]$, for each index $i \in I$, define the „biproduct“ $(F \oplus G)(i) := F(i) \oplus G(i)$, which it naturally associates with projections $\pi_{F,i} : (F \oplus G)(i) \twoheadrightarrow F(i)$ and $\pi_{G,i} : (F \oplus G)(i) \twoheadrightarrow G(i)$, also with inclusions $\iota_{F,i} : F(i) \hookrightarrow (F \oplus G)(i)$ and $\iota_{G,i} : G(i) \hookrightarrow (F \oplus G)(i)$.

Now, we claim for every $i \rightarrow j$ in I , there exists a natural morphism $h_{ij} : (F \oplus G)(i) \rightarrow (F \oplus G)(j)$. Let's consider the following diagram:

$$\begin{array}{ccccc}
 F(i) & \xleftarrow{\pi_{F,i}} & (F \oplus G)(i) & \xrightarrow{\pi_{G,i}} & G(i) \\
 f_{ij} \downarrow & & \downarrow \exists! h_{ij} & & \downarrow g_{ij} \\
 F(j) & \xleftarrow{\pi_{F,j}} & (F \oplus G)(j) & \xrightarrow{\pi_{G,j}} & G(j)
 \end{array}$$

Where, the uniqueness of h_{ij} is given by the two maps $f_{ij} \circ \pi_{F,i} : (F \oplus G)(i) \rightarrow F(j)$, $g_{ij} \circ \pi_{G,i} : (F \oplus G)(i) \rightarrow G(j)$, and the fact that $(F \oplus G)(j)$ (with the two projections) is a product of $F(j), G(j)$ in $\mathbf{Ab}$.

Then, the association $F \oplus G : I \rightarrow \mathbf{Ab}$ by $i \mapsto (F \oplus G)(i)$, and $F \oplus G(i \rightarrow j) = h_{ij}$ is a functor by the universality of product.

(**Note:** If using the inclusions as the proof, we'll still get the same morphism, as the biproduct property guarantees the h_{ij} to agree, regardless if we use projections or inclusions to derive it). Also, the „projections“ and „inclusions“ component wise clearly form a naturality square, showing $\pi_F : F \oplus G \rightarrow F$ (resp. π_G for G) and $\iota_F : F \rightarrow F \oplus G$ (resp. ι_G for G) are all morphisms in $[I, \mathbf{Ab}]$.

Finally, we claim the universality of $F \oplus G$ as a biproduct (which we'll prove the product part, as the coproduct part is with similar proof).

Consider another direct system L together with two morphisms $p_F : L \rightarrow F$ and $p_G : L \rightarrow G$, then notice that component wise, the universality of $(F \oplus G)(i)$ as product of $F(i), G(i)$ guarantees a unique morphism $p_i : L(i) \rightarrow (F \oplus G)(i)$ as follow:

$$\begin{array}{ccccc}
 & & L(i) & & \\
 & \swarrow p_{F,i} & \downarrow \exists! p_i & \searrow p_{G,i} & \\
 F(i) & \xleftarrow{\pi_{F,i}} & (F \oplus G)(i) & \xrightarrow{\pi_{G,i}} & G(i)
 \end{array}$$

Which, given any morphism $i \rightarrow j$, one has the following large „commutative tent“:

$$\begin{array}{ccccc}
 & & L(i) & \xrightarrow{l_{ij}} & L(j) \\
 & \swarrow p_{F,i} & \downarrow p_i & \searrow p_{F,j} & \\
 F(i) & \xrightarrow{f_{ij}} & F(j) & & \\
 & \swarrow \pi_{F,i} & \downarrow p_{G,i} & \searrow \pi_{F,j} & \\
 & & (F \oplus G)(i) & \xrightarrow{h_{ij}} & (F \oplus G)(j) \\
 & \swarrow \pi_{G,i} & \downarrow p_{G,i} & \searrow \pi_{G,j} & \\
 & & G(i) & \xrightarrow{g_{ij}} & G(j)
 \end{array}$$

Which, notice the middle „verticle square“ is also a naturality square, based on the following observation:

$$\begin{aligned}
 \pi_{G,j} \circ (h_{ij} \circ p_i) &= g_{ij} \circ (\pi_{G,i} \circ p_i) = g_{ij} \circ p_{G,i} \\
 &= p_{G,j} \circ l_{ij} = \pi_{G,j} \circ (p_j \circ l_{ij})
 \end{aligned} \tag{1.2}$$

Repeating similar proof for the other side of the tent, we also have the following:

$$\pi_{F,j} \circ (h_{ij} \circ p_i) = \pi_{F,j} \circ (p_j \circ l_{ij}) \tag{1.3}$$

Where this is based on the fact that p_F, p_G are natural transformations, and the product property of morphisms h_{ij} and p_i, p_j .

Also, consider the fact that the morphism $p_{F,j} \circ l_{ij} : L(i) \rightarrow F(j)$ and $p_{G,j} \circ l_{ij} : L(i) \rightarrow G(j)$, they factor uniquely through $(F \oplus G)(j)$, and the unique factored morphism has composition with the projections that recovers the two listed morphisms.

Then, based on (1.2), (1.3) two equations, we know that $p_j \circ l_{ij}, (h_{ij} \circ p_i) : L(i) \rightarrow (F \oplus G)(j)$ both satisfy such requirement (since $\pi_{F,j} \circ (p_j \circ l_{ij}) = p_{F,j} \circ l_{ij}$, and $\pi_{G,j} \circ (p_j \circ l_{ij}) = p_{G,j} \circ l_{ij}$). Hence, this enforces $p_j \circ l_{ij} = h_{ij} \circ p_i$ using the universality of product. This proves the middle is a naturality square, or $p : L \rightarrow F \oplus G$ is well-defined.

Finally, the morphism p is unique, simply because component wise it must be realized as a product morphism, hence must be unique. This proves the universality of $F \oplus G$ as a product in $[I, \mathbf{Ab}]$. (Note: Repeat the proof with inclusions, and reverse the tent provides the case for coproduct :)

Using similar proof, one can also construct arbitrary small product and coproduct in similar sense (except over infinite direct systems they'll be different).

So, with all these conditions, $[I, \mathbf{Ab}]$ is an abelian category. $\square$

2 Direct Limits and Exactness

Given the Functor Category $[I, \mathbf{Ab}]$, all direct system $F \in [I, \mathbf{Ab}]$ has direct limit $\text{colim } F$ (well-defined up to isomorphism), in particular we'll choose this to be the classical construction through direct sums and quotients. Now, we claim that similar ideas can be done on the morphisms:

Proposition

Given morphism $\varphi : F \rightarrow G$ in $[I, \mathbf{Ab}]$, it descends to a homomorphism of direct limits, denoted as $\bar{\varphi} : \text{colim } F \rightarrow \text{colim } G$.

Proof: Notice that every $i \in I$ corresponds to a morphism $\varphi_i : F(i) \rightarrow G(i)$, composing with each inclusion $\iota_{G,i} : G(i) \hookrightarrow \bigoplus_{i \in I} G(i)$, one gathers a family of map $(\iota_G \circ \varphi)_i : F(i) \rightarrow \bigoplus_{i \in I} G(i)$. Hence, using the universality of biproduct, one gets $\bigoplus \varphi : \bigoplus_{i \in I} F(i) \rightarrow \bigoplus_{i \in I} G(i)$ (intuitively, it's a component-wise map, each being the φ_i).

Now, given the direct limit is constructed as a suitable quotient, one has projection $\pi_F : \bigoplus_{i \in I} F(i) \twoheadrightarrow \text{colim } F$ and $\pi_G : \bigoplus_{i \in I} G(i) \twoheadrightarrow \text{colim } G$. Then, the composition of morphisms give the morphism $\pi_G \circ (\bigoplus \varphi) : \bigoplus_{i \in I} F(i) \rightarrow \text{colim } G$. The goal is to prove this uniquely factors through $\text{colim } F$.

Let's recall the equivalence relation for direct limit: Given $a_i \in F(i)$ and $a_j \in F(j)$, one has $a_i \sim a_j$ iff there exists an upper bound $k \geq i, j$, such that $f_{ik}(a_i) = f_{jk}(a_j)$ in $F(k)$.

However, if $a_i \sim a_j$, then notice that $\varphi_i(a_i) \in G(i)$ and $\varphi_j(a_j) \in G(j)$ satisfies the following:

$$g_{ik} \circ \varphi_i(a_i) = \varphi_k \circ f_{ik}(a_i) = \varphi_k \circ f_{jk}(a_j) = g_{jk} \circ \varphi_j(a_j) \quad (2.1)$$

This shows that $\varphi_i(a_i) \sim \varphi_j(a_j)$ in direct system G , or $\pi_G(\varphi_i(a_i) - \varphi_j(a_j)) = 0$.

As a result, since $(\bigoplus \varphi)(a_i - a_j) = \varphi_i(a_i) - \varphi_j(a_j)$ (since it's component-wise morphism), so $\pi_G \circ (\bigoplus \varphi)(a_i - a_j) = 0$, showing the subgroup of equivalence relation in direct system F is inside the kernel of the map.

Hence, this uniquely factors through the quotient of $\bigoplus_{i \in I} F(i)$ with the equivalence relation, namely the direct limit $\text{colim } F$, which can be denoted as a morphism $\bar{\varphi} : \text{colim } F \rightarrow \text{colim } G$, such that $\pi_G \circ (\bigoplus \varphi) = \bar{\varphi} \circ \pi_F$. $\square$

As a remark, this assignment $(\varphi : F \rightarrow G) \mapsto (\bar{\varphi} : \text{colim } F \rightarrow \text{colim } G)$ is functorial (because the composition of homomorphisms between direct sums of elements in the direct system, is equivalent to compose the morphisms between direct systems first, then take the homomorphism between the direct sums). Hence, this generates a functor $D : [I, \mathbf{Ab}] \rightarrow \mathbf{Ab}$, by $F \mapsto \text{colim } F$, and morphism $(\varphi : F \rightarrow G) \mapsto (\bar{\varphi} : \text{colim } F \rightarrow \text{colim } G)$.

Now, we wish to claim something even stronger, that D not only is a functor, but also preserves additive structure, and exactness.

Theorem

$D : [I, \mathbf{Ab}] \rightarrow \mathbf{Ab}$ is an additive functor between abelian categories.

Proof: There are two ways to approach: Either proving that D preserves biproduct (the hard way), or do an element-wise proof. We'll choose the latter (simply because I'm lazy :)

Recall that given morphism $\varphi : F \rightarrow G$, one has induced homomorphism $(\bigoplus \varphi) : \bigoplus_{i \in I} F(i) \rightarrow \bigoplus_{i \in I} G(i)$, and the induced homomorphism on direct limits $\bar{\varphi} : \text{colim } F \rightarrow \text{colim } G$ that satisfy the following commutative diagram:

$$\begin{array}{ccc}
\bigoplus_{i \in I} F(i) & \xrightarrow{\oplus \varphi} & \bigoplus_{i \in I} G(i) \\
\pi_F \downarrow & & \downarrow \pi_G \\
\operatorname{colim} F & \xrightarrow{\overline{\varphi}} & \operatorname{colim} G
\end{array}$$

Now, given any $\varphi, \psi : F \rightarrow G$, notice that $\varphi + \psi$ is a morphism that has addition component wise. Hence, the general homomorphism $(\oplus (\varphi + \psi)) : \bigoplus_{i \in I} F(i) \rightarrow \bigoplus_{i \in I} G(i)$ in fact equals the addition $(\oplus \varphi) + (\oplus \psi)$ (since the morphism generated between coproducts preserve the component wise addition). As a result, one has the following:

$$\begin{aligned}
\overline{\varphi + \psi} \circ \pi_F &= \pi_G \circ (\oplus (\varphi + \psi)) = \pi_G \circ (\oplus \varphi + \oplus \psi) \\
&= \pi_G \circ (\oplus \varphi) + \pi_G \circ (\oplus \psi) = \overline{\varphi} \circ \pi_F + \overline{\psi} \circ \pi_F \\
&= (\overline{\varphi} + \overline{\psi}) \circ \pi_F
\end{aligned} \tag{2.2}$$

Then, using the fact that π_F is an epimorphism (which is right-cancellative), one guarantees $\overline{\varphi + \psi} = \overline{\varphi} + \overline{\psi}$. Hence, $D(\varphi + \psi) = D\varphi + D\psi$, proving the additive property of D .

□

And, here is the claim on exactness:

Theorem

$D : [I, \mathbf{Ab}] \rightarrow \mathbf{Ab}$ as an additive function, is exact.

Proof: Here, we'll also try and work with element-wise proof (the general categorical claim is too hard to be honest). Here, let's fix two morphisms $\varphi : F \rightarrow G$ and $\psi : G \rightarrow H$ in $[I, \mathbf{Ab}]$. We'll still proceed with several smaller claims:

I. Equivalence of Exactness and Component-Wise Exactness:

For detail, we claim that the sequence $0 \rightarrow F \rightarrow G \rightarrow H \rightarrow 0$ is exact, iff every index $i \in I$ has the homomorphisms $\varphi_i : F(i) \rightarrow G(i)$ and $\psi_i : G(i) \rightarrow H(i)$ satisfy $0 \rightarrow F(i) \rightarrow G(i) \rightarrow H(i) \rightarrow 0$ being exact.

Notice that $0 \rightarrow F \rightarrow G \rightarrow H \rightarrow 0$ is exact $\iff \varphi$ is monomorphism, ψ is epimorphism, and the morphisms $\operatorname{im}(\varphi) = \ker(\psi)$.

However, notice that all of these are equivalent to component-wise properties:

- φ is monomorphic $\iff$ each φ_i is monomorphic.
- ψ is epimorphic $\iff$ each ψ_i is epimorphic.
- $\operatorname{im}(\varphi) = \ker(\psi) \iff$ each $\operatorname{im}(\varphi_i) = \ker(\psi_i)$.

Which, all three conditions are equivalent to $0 \rightarrow F(i) \rightarrow G(i) \rightarrow H(i) \rightarrow 0$ being exact for all $i \in I$.

II. Equivalence of Component-Wise Exactness and Direct Sums' Exactness:

(**Remark:** This statement is not true in any abelian category, only in Grothendieck Categories).

For detail, we claim that each index $i \in I$ has $0 \rightarrow F(i) \rightarrow G(i) \rightarrow H(i) \rightarrow 0$ being exact $\iff$ the gathered homomorphisms $(\oplus \varphi) : \bigoplus_{i \in I} F(i) \rightarrow \bigoplus_{i \in I} G(i)$ and $(\oplus \psi) : \bigoplus_{i \in I} G(i) \rightarrow \bigoplus_{i \in I} H(i)$ satisfy $0 \rightarrow \bigoplus_{i \in I} F(i) \rightarrow \bigoplus_{i \in I} G(i) \rightarrow \bigoplus_{i \in I} H(i) \rightarrow 0$ being exact.

First, about $\oplus \varphi$ being monomorphic, the set-theoretic side guarantees that if $(\oplus \varphi)((a_i)_{i \in I}) = (\varphi_i(a_i))_{i \in I} = 0$ iff each $\varphi_i(a_i) = 0$ iff each $a_i \in \ker(\varphi_i)$. So, $\ker(\oplus \varphi) = \bigoplus_{i \in I} \ker(\varphi_i)$. Hence, $\oplus \varphi$ is a

monomorphism $\iff$ each direct summand (or each $\ker(\varphi_i)$) is 0, which is equivalent to component-wise monomorphism. This takes care of the left most part.

Then, about $\oplus \psi$ being epimorphic, again set-theoretic side guarantees that if each ψ_i is surjective, then $\oplus \psi$ is surjective (as all element $(b_i)_{i \in I}$ in $\oplus_{i \in I} H(i)$, each has some $a_i \in G(i)$, where $\psi_i(a_i) = b_i$, so $(\oplus \psi)((a_i)_{i \in I}) = (\psi_i(a_i))_{i \in I} = (b_i)_{i \in I}$; this is well-defined because only finitely many $b_i \neq 0$, so only finitely many a_i needs to be chosen as nonzero). On the other hand, if $\oplus \psi$ is surjective, then each ψ_i must also be surjective (since looking at any element $a_i \in H(i) \hookrightarrow \oplus_{i \in I} H(i)$, it is mapped to by some $(a_i)_{i \in I}$ in $\oplus_{i \in I} G(i)$, so in particular one has $\psi_i(a_i) = b_i$). Hence, the epimorphism part is also equivalent.

Finally, about the middle part, since one has $\ker(\oplus \psi) = \oplus_{i \in I} \ker(\psi_i)$, and $\text{im}(\oplus \varphi) = \oplus_{i \in I} \text{im}(\varphi_i)$, then the two are equal $\iff$ each $\text{im}(\varphi_i) = \ker(\psi_i)$, happens iff it's exact at the middle of every component.

This finishes the proof of the equivalence.

III. Direct Sum Exactness implies Direct Limit Exactness:

Now, we know $0 \rightarrow \oplus_{i \in I} F(i) \rightarrow \oplus_{i \in I} G(i) \rightarrow \oplus_{i \in I} H(i) \rightarrow 0$ is exact (because $0 \rightarrow F \rightarrow G \rightarrow H \rightarrow 0$ is, and the previous two parts). Let's conclude it for the direct limits:

1. Given $(\oplus \psi) : \oplus_{i \in I} G(i) \twoheadrightarrow \oplus_{i \in I} H(i)$ being surjective, consider this commutative diagram:

$$\begin{array}{ccc} \oplus_{i \in I} G(i) & \xrightarrow{\oplus \psi} & \oplus_{i \in I} H(i) \\ \pi_G \downarrow & & \downarrow \pi_H \\ \text{colim } G & \xrightarrow{\bar{\psi}} & \text{colim } H \end{array}$$

Which, $\bar{\psi} \circ \pi_G = \pi_H \circ (\oplus \psi)$, while the latter is surjective, this enforces $\bar{\psi}$ to be surjective.

2. Given $(\oplus \varphi) : \oplus_{i \in I} F(i) \hookrightarrow \oplus_{i \in I} G(i)$ being injective, consider the commutative diagram:

$$\begin{array}{ccc} \oplus_{i \in I} F(i) & \xhookrightarrow{\oplus \varphi} & \oplus_{i \in I} G(i) \\ \pi_F \downarrow & & \downarrow \pi_G \\ \text{colim } F & \xrightarrow{\bar{\varphi}} & \text{colim } G \end{array}$$

Which, given any $\bar{a} \in \ker(\bar{\varphi})$, take a representative $a \in \oplus_{i \in I} F(i)$, then one has $\bar{\varphi} \circ \pi_F(a) = \bar{\varphi}(\bar{a}) = 0$, so $\pi_G \circ (\oplus \varphi)(a) = \bar{\varphi}(a) = 0$, showing $(\oplus \varphi)(a)$ is inside the subgroup generated by the equivalence relation in G .

Yet, notice that this enforces a to also be in the subgroup generated by the equivalence relation in F (which we'll check that the relations in F must be precisely carried over to G , no new relations can be created in the image):

If a_i, b_j in F satisfies $\varphi_i(a_i) \sim \varphi_j(b_j)$ in G , then one has some upper bound $k \geq i, j$ in I , such that $g_{ik} \circ \varphi_i(a_i) = g_{jk} \circ \varphi_j(b_j)$, using the naturality of φ , one gets $\varphi_k \circ f_{ik}(a_i) = \varphi_k \circ f_{jk}(b_j)$, then the injectivity of φ_k implies $f_{ik}(a_i) = f_{jk}(b_j)$, or $a_i \sim a_j$.

Hence, if $(\oplus \varphi)(a)$ lives in the subgroup generated by the equivalence relation of G , its preimage (namely a only by injectivity) must live in the subgroup generated by the equivalence relation of F . Hence, one has $\pi_F(a) = \bar{a} = 0$, proving the injectivity of $\bar{\varphi}$.

3. Since the big commutative diagram commutes (i.e. $\bar{\psi} \circ \bar{\varphi} \circ \pi_F = \bar{\psi} \circ \pi_G \circ (\oplus \varphi) = \pi_H \circ (\oplus \psi) \circ (\oplus \varphi) = 0$, with π_F being epimorphic, then $\bar{\psi} \circ \bar{\varphi} = 0$).

On the other hand, any $\bar{b} \in \ker(\bar{\psi})$ has its representative $b \in \oplus_{i \in I} G(i)$ satisfies $0 = \bar{\psi}(\bar{b}) = \bar{\psi} \circ \pi_G(b) = \pi_H \circ (\oplus \psi)(b)$

□